Classifiers

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Introduction to Classifiers

- Definition: A classifier is an algorithm that maps input data (features) into specific categories or "classes."
- The Goal: To predict the class label for new, unseen data based on patterns learned from training data.
- Key Components:
- Feature Vector: The numerical representation of the object.
- Decision Boundary: The line or surface that separates different classes.

The Pattern Recognition Pipeline

Before classification happens, data must move through a standard workflow:

1. Data Acquisition: Gathering raw data (images, signals, text).
2. Preprocessing: Noise removal and normalization.
3. Feature Extraction: Identifying the most relevant characteristics (e.g., edges, textures, or statistical means).
4. Classification: Applying the model to assign a label.
5. Post-processing: Refining the results (e.g., error correction).

Types of Classification Learning

Classifiers are generally categorized by how they learn from data:

- Supervised Learning: The model is trained on a labeled dataset (the "answer key" is provided).
  - Examples: Identifying spam emails, medical diagnosis.
- Unsupervised Learning: The model finds hidden patterns or clusters in unlabeled data.
  - Examples: Customer segmentation, image compression.
- Semi-Supervised Learning: Uses a small amount of labeled data and a large amount of unlabeled data.

Linear vs. Non-Linear Classifiers

- Linear Classifiers: Separate classes using a straight line (in 2D) or a hyperplane (in higher dimensions).
- Best for: Linearly separable data.
- Examples: Logistic Regression, Linear SVM.
- Non-Linear Classifiers: Use curved or complex boundaries to separate data that cannot be split by a straight line.
- Examples: K-Nearest Neighbors (KNN), Neural Networks, Kernel SVM.

Common Classification Algorithms (Part 1)

1. Naive Bayes
   - Based on Bayes' Theorem.
   - Assumes "naively" that features are independent.
   - Pros: Fast and efficient for text classification.
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2. K-Nearest Neighbors (KNN)
   - A "lazy learner" that classifies a point based on the majority class of its k closest neighbors.
   - Uses distance metrics like Euclidean distance.

Common Classification Algorithms (Part 2)

3. Support Vector Machines (SVM)
   - Finds the "Optimal Hyperplane" that maximizes the margin between two classes.
   - Uses Kernels to handle non-linear data.
4. Decision Trees & Random Forests
   - Decision Tree: A flowchart-like structure using branching logic.
   - Random Forest: An ensemble method that combines multiple decision trees for higher accuracy.

Evaluation Metrics

How do we know if our classifier is "good"?

- Confusion Matrix: A table showing True Positives, True Negatives, False Positives, and False Negatives.
- Accuracy: Overall percentage of correct predictions.
- Precision & Recall: Critical for imbalanced datasets (e.g., fraud detection).
- F1-Score: The harmonic mean of Precision and Recall.

Challenges in Classification

- Overfitting: The model learns the noise in training data and performs poorly on new data.
- Underfitting: The model is too simple to capture the underlying pattern.
- Dimensionality: Too many features can lead to the "Curse of Dimensionality," making classification computationally expensive.

Summary & Conclusion

- Summary: Classification turns raw data into intelligence. Choosing the right algorithm depends on the data size and complexity.
- Conclusion: Classifiers are the engine of AI. While simple models like Naive Bayes are fast, complex models like Random Forests or Neural Networks provide higher accuracy for difficult tasks.

References

- Pattern Classification by Duda, Hart, and Stork.
- Pattern Recognition and Machine Learning by Christopher Bishop.
- Online documentation: Scikit-learn.org.

Thank you